

UNIT – V - NANO DEVICES

- Electron Density in Bulk Material
- Quantum Confinement
- Single Electron Transistor - SET
- Carbon Nanotubes (CNTs)
- Quantum Dot laser (QDL)

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1. Electron density in bulk material

One of the major ways in which small volume materials differ from bulk materials, is the number of available energy states.

Given as

$$N = \left(\frac{8\pi}{3} \right) (2mE)^{\frac{3}{2}} \left(\frac{a^3}{h^3} \right) \quad \dots\dots\dots (1)$$

Number of energy states per unit volume,

$$n = \frac{N}{a^3} \left(\frac{8\pi}{3} \right) \frac{(2mE)^{\frac{3}{2}}}{h^3} \quad \dots\dots\dots (2)$$

where,

n - the number of energy states/volume

E - maximum energy level

m - mass of an electron

h - Planck's constant.

Density of states function Z(E) is obtained after taking the derivative of this expression w.r.to $\frac{dn}{dE}$

ie., no. of available electron energy states per unit volume per unit energy

$$\begin{aligned} Z(E) &= \frac{dn}{dE} \\ &= \frac{d}{dE} \left(\frac{8\pi}{3} \frac{(2mE)^{\frac{3}{2}}}{h^3} \right) \\ &= \frac{8\pi}{3} \frac{(2m)^{\frac{3}{2}}}{h^3} \frac{dE^{\frac{3}{2}}}{dE} \\ &= \frac{8\pi}{3} \frac{(2)(2)^{\frac{1}{2}} m^{\frac{3}{2}}}{h^3} \left(\frac{3}{2} E^{\frac{1}{2}} \right) \\ Z(E) &= \frac{8\pi \sqrt{2} (m)^{\frac{3}{2}}}{h^3} \sqrt{E} \quad \dots\dots\dots (3) \end{aligned}$$

$$\therefore n_e = \frac{8\pi}{3} \left(\frac{(2mE_F)^{\frac{3}{2}}}{h^3} \right) \quad \dots\dots\dots (4)$$

where n_e – electron density at 0 K.

Size dependence on fermi energy

Fermi energy

It is defined as the highest energy level occupied by the electron at 0 K in a metal.
From eqn (4)

$$E_F = \frac{h^2}{2m} \left(\frac{3n_e}{8\pi} \right)^{\frac{2}{3}}$$

Here n is only variable and remaining are constants.

As the electron density is a property of the material, Fermi energy does not vary with material size. The average spacing between energy states is inversely proportional to the volume of the solid.

$$\Delta E \propto \frac{1}{a^3}$$

2. Quantum confinement

The effect achieved by reducing the volume of a solid so that the energy levels within it become discrete is called quantum confinement. ie., if we decrease the particle to nano size, the decrease in confining size creates the energy level discrete and the band gap energy increases and the discrete energy levels also increases. Due to this significant changes occur in electrical and optical materials.

Quantum structures

When a bulk material is reduced in its size at least one of its dimensions in the order of few nanometers, then it is known as Quantum structure.

Based on the confinement direction, a quantum confined structure will be classified as follows.

STRUCTURE	QUANTUM CONFINEMENT	NO. OF DIMENSIONS
Bulk	0	3(x, y, z)
Quantum well	1(z)	2(x, y)
Quantum wire	2(x, y)	1(z)
Quantum dot	3(x, y, z)	0

Density of energy states in 3 dimension

We know that for 3 dimensions

$$Z(E) = \frac{\pi}{2} \left(\frac{8m^*}{(2m\hbar)^2} \right)^{\frac{3}{2}} E^{\frac{1}{2}}$$

$$= \frac{\pi}{2} \left(\frac{2m^*}{\pi^2 \hbar^2} \right)^{\frac{3}{2}} E^{\frac{1}{2}}$$

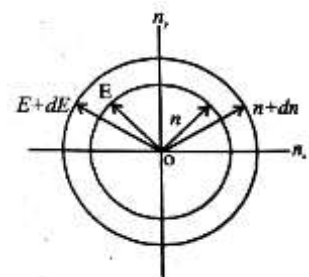
$$Z(E)^{3D} \propto E^{\frac{1}{2}}$$

Density of states in quantum well (2 dimension)

The two dimensional density of energy states is the number of states per unit energy.

Consider the electron in 2-dimensional bounded region of space. We need to find how many quantum states lie within a boundary energy i.e. between E & E+dE.

Now reduced phase space consists only x & y plane.



i.e.,
$$n^2 = n_x^2 + n_y^2$$

Number of energy states within circle of radius 'n' = $\frac{1}{4}\pi n^2$

(1/4th of the circle has the quantum intervals)

Similarly

Number of energy states within circle of radius n+dn = $\frac{1}{4}\pi(n+dn)^2$

$\therefore$ Number of energy states between n & n+dn = $\frac{1}{4}\pi(n^2 + 2ndn + dn^2 - n^2)$

Neglecting higher powers dn² we have

$$\begin{aligned} N(E) &= \pi/4(2ndn) \\ &= \frac{\pi}{2}(ndn) \end{aligned} \quad \dots\dots (3)$$

We know that,

$$E = \frac{n^2 h^2}{8ma^2} \quad \dots\dots (4)$$

$$ndn = \frac{8ma^2 dE}{2h^2} \quad \dots\dots (5)$$

Substituting (5) in (3) and applying Pauli's exclusion principle

$$\begin{aligned} \therefore N(E)dE &= \frac{\pi}{2} \left(\frac{8m^*}{2h^2} \right) a^2 dE \\ &= \frac{\pi}{4} \left(\frac{8m^*}{h^2} \right) a^2 dE \end{aligned} \quad \dots\dots (6)$$

$$\begin{aligned} Z(E) &= \frac{N(E)dE}{a^2} \\ Z(E)dE &= \frac{\frac{\pi}{4} \left(\frac{8m^*}{h^2} \right) a^2 dE}{a^2} \end{aligned}$$

$$Z(E) = \frac{\pi}{2} \left(\frac{8m^*}{(2\pi\hbar)^2} \right)$$

$$\begin{aligned} \therefore \hbar^2 &= \left(\frac{h}{2\pi} \right)^2 \\ \therefore h^2 &= (2\pi\hbar)^2 \end{aligned}$$

$$Z(E) = \frac{m}{\pi\hbar^2}$$

for $E \geq E_0$

where E_0 is the ground state energy level of quantum well.

$$Z(E)^{2D} = \frac{m^*}{\pi\hbar^2} \sum_n \sigma(E - E_n)$$

Where E_n are energies of quantized states and $\sigma(E - E_n)$ is step function.

$$Z(E)^{2D} \propto E^0 = \text{constant}$$

For energy $E \geq E_0$, the 2D density of states is constant and does not depend on energy.

Density of states in quantum wire (1 dimension)

Consider the 1D system, the quantum wire in which only one direction of motion is allowed.

In one dimension, such as for a quantum wire, the density of energy states is the no. of available energy states per unit length per unit energy around an energy E .

Let the electron inside the wire be confined in 1D infinite well with zero potential inside and infinite potential outside the wire.

At $x = 0$: $V(x) = 0$ for electron inside the wire

At $x = a$: $V(x) = \infty$ for electron outside the wire. Now reduced phase is

$$n^2 = n_x^2 \dots\dots\dots(1)$$



The no of energy states lying in the interval of length

$$N(E)dE = n + dn - n$$

$$N(E)dE = dn \dots\dots\dots(2)$$

$$E = \frac{n^2 h^2}{8ma^2}$$

We know that,

$$n = \left(\frac{8m^* a^2 E}{h^2} \right)^{\frac{1}{2}}$$

$$= \left(\frac{8m^*}{h^2} \right)^{\frac{1}{2}} a E^{\frac{1}{2}}$$

$$dn = \left(\frac{8m^*}{h^2} \right)^{\frac{1}{2}} a \left(\frac{1}{2} \right) E^{-\frac{1}{2}} dE \dots\dots\dots(3)$$

Substitute (3) in (2)

$$N(E)dE = \left(\frac{8m^*}{h^2} \right)^{\frac{1}{2}} a \left(\frac{1}{2} \right) E^{-\frac{1}{2}} dE \dots\dots\dots(4)$$

Applying Pauli's exclusion principle in (4),

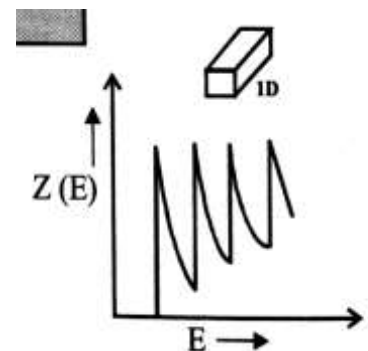
$$N(E)dE = 2 \left(\frac{8m^*}{h^2} \right)^{\frac{1}{2}} a \left(\frac{1}{2} \right) E^{-\frac{1}{2}} dE \dots\dots\dots(5)$$

$$Z(E)dE = \frac{N(E)dE}{a}$$

$$Z(E) = \frac{8m^*}{(2\pi\hbar)^2} E^{\frac{1}{2}} \dots\dots\dots(6)$$

$$Z(E)^{1D} = \frac{1}{\pi\hbar} \sqrt{\frac{2m^*}{(E - E_0)}} \quad \text{for } E \geq E_0 \dots\dots\dots(7)$$

It has functional dependence on energy for more than 1 state it is given as



$$Z(E)^{1D} = \frac{1}{\pi\hbar} \sum_n \sqrt{\frac{2m^*}{(E - E_n)}} \sigma(E - E_n)$$

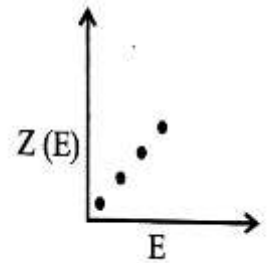
Density of states in quantum dot(0-dimension)

In a zero – dimension system ie., quantum dot is confined in all 3D and hence no motion of electron is possible. So each state of energy level is occupied by 2 electrons. So density of energy states is a delta function is given by

$$Z(E)^{0D} = 2\sqrt{(E - E_n)}$$

Factor 2 is for spin. For more than 1 quantum state

$$Z(E)^{0D} = \sum_n 2\delta\sqrt{(E - E_n)}$$



3.Single electron phenomena

Transistor operations, such as amplifiers, switches, electrometers, oscillators are operated using single electron or Quantum dot, this is known as single electron phenomena.

To get this single electron phenomena the following conditions are followed:

- 1) Single electron or Quantum dot should be kept isolated.
- 2) No addition of electron or deletion of electron to quantum dot.
- 3) There should be no tunneling.

These conditions can be achieved by coulomb blockade effect.

Coulomb blockade effect

The prohibition (or) suppression of tunneling is called Coulomb blockade effect.

We know that Coulomb force is electrostatic, so when two charges are near one another they exert Coulomb forces upon each other. If two charges are same the force is repulsive.

In quantum dot, the charges are all negative electrons trying to bring them forcefully together create Coulomb force, so it repels.

This is coulomb blockade effect, it presents tunneling to and form a quantum dot. This effect is measured as.

$$C_{dot} = Gd \quad \dots\dots\dots (1)$$

Where, C_{dot} – capacitance of quantum dot ie., (measure of how much electric charge it can store)

ϵ – permittivity, d - diameter of the dot, G - geometrical form (for disk like quantum dot $G = 4$, for spherical $G = 2\pi$).

Therefore the energy required to add negatively charged electron to the dot is

$$E_c = \frac{e^2}{2C_{dot}} \quad \dots\dots\dots (2)$$

where, e = charge of electron.

$$\begin{aligned} \because q &= CV \\ e &= CV \\ \therefore v &= \frac{e}{C} \\ E &= \frac{1}{2} CV^2 = \frac{1}{2} \frac{e^2}{C} \\ \therefore E_c &= \frac{e^2}{2C} \end{aligned}$$

From eqn (2) E_c is inversely proportional to C_{dot} .
ie., if capacitance is large it can accommodate another electron easily.
But if it is small it can block tunneling.

Coulomb blockade can measure unwanted tunneling when its energy is greater than the thermal energy of the electron
ie., $E_c \gg K_B T$

This is achievable when the dot is very smaller.

Single Electron Transistor (SET)

Definition

A transistor made from a quantum dot that controls the current from source to drain one electron at a time is called SET.

Construction

It is similar to conventional FET, the difference is that instead of PN junction, we have tunneling junction and of channel we have quantum dot.

Gate electrode biasing controls tunneling and biasing of source and drain controls current flow direction.

The gate bias voltage must be greater to overcome coulomb blockade energy.

If E is the energy required to move charge Q across potential difference V then

$$E = VQ \quad \dots\dots\dots (1)$$

$$Q = e$$

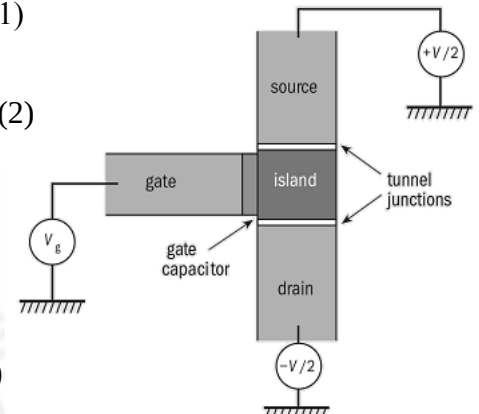
$$\therefore V = \frac{E}{e} \text{ (or) } \frac{E_c}{e} \quad \dots\dots\dots (2)$$

But

$$E = E_c \quad \& \quad E_c = \frac{e^2}{2C}$$

But

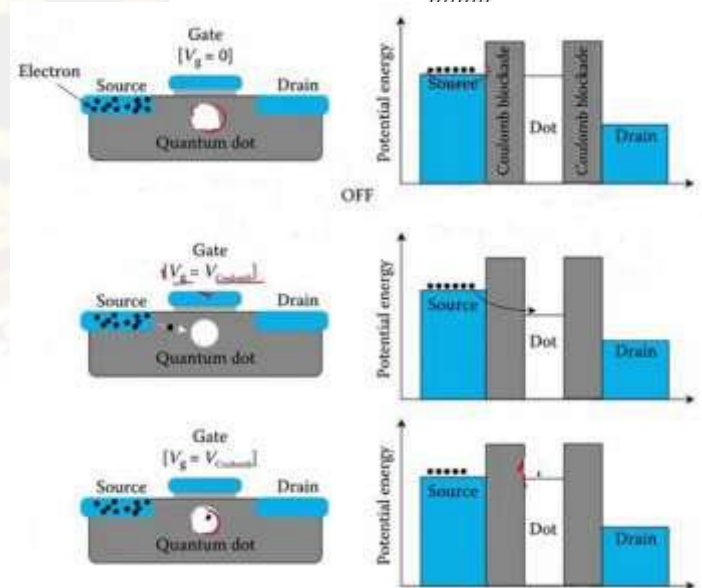
$$\therefore V = \frac{\frac{e^2}{2C}}{e} \Rightarrow V_g = \frac{e}{2C} \quad \dots\dots\dots (3)$$



This is the voltage required to overcome E_c .

Working:

- 1) When SET is in OFF mode the corresponding P.E diagram shows that it is not energetically favorable for electrons in the source to tunnel via dot.
- 2) When it is in ON mode at the lowest setting electron tunnel one at a time from source to drain.
- 3) This is made possible by first applying the proper gate voltage $V_{gate} = e/2C_{dot}$, so that the potential energy of the dot is made low enough to encourage an electron tunnel through it.
- 4) Once the electron enters dot, its potential energy increases.
- 5) With dot empty and potential lower again, the process is repeated thus number of electrons are connected using dot. Thus the ON and OFF condition is used to make an effective switch.



Advantages

- 1) Fast information transfer.
- 2) Suitable for high density memory.
- 3) Can be used for quantum computer.

Disadvantages

- 1) Size of the dot should be very less to operate it under room temperature.
- 2) Very hard to fabricate.

Applications

- 1) Used for mass data storage.

- 2) Used in sensor technology.
- 3) Designing of AND and NOR Gate.
- 4) Used in sensitive electrometer.
- 5) Used in fabrication of homo-dyne receiver.

Ballistic transport

Definition

When the mean free path of the electron is longer than the dimension of the medium through which the electron travels is called Ballistic transport.

Explanation

When the length 'L' of the conductor becomes much smaller than the mean free path 'L_m' the transport is termed 'Ballistic' meaning that the electrons do not scatter during the time it takes to travel through the conductor.

For example, ballistic transport can be observed in a metal nanowire. This is because the wire is of the size of a nanometer and the mean free path can be longer than in a metal.

Condition for ballistic transport

The mean free path can be increased by reducing the number of impurities in a crystal (or) by lowering its temperature.

$$(i) \quad L \ll L_m \text{ and}$$

$$(ii) \quad L \ll L_\phi$$

where

L – length of the conductor

L_m – mean free path

L_φ - length over which an electron can travel before having an inelastic collision.

This is also called the length over which an electron wave function retains its coherence.

For $L \ll L_m$ and $L \ll L_\phi$ we have Ballistic transport.

The electron doesn't hit anything as it travels through the material, and therefore, there is no momentum. Thus the electron's wave function can be obtained from Schrodinger's equation.

Applications

- 1) Ultra – short – channel semiconducting FETS.
- 2) Carbon nanotube transistors.

4. Carbon nanotubes (CNTs)

Definition

Carbon nanotubes (CNTs) are molecular-scale tubes of graphitic carbon with outstanding properties. They are among the stiffest and strongest fibers researched till date, with remarkable electronics properties and applications.

The hexagonal lattice of carbon is graphite.

A single layer of graphite is called graphene.

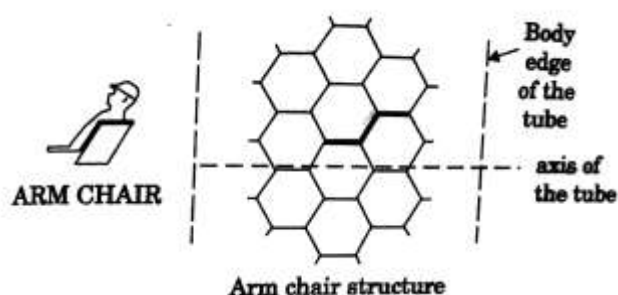
Types of CNTs:

1. Based on rolling a graphite sheet - Arm Chair, Zig Zag and Chiral
2. Based on number of layers - Single walled nanotubes and Multi walled nanotubes

Based on rolling a graphite sheet:

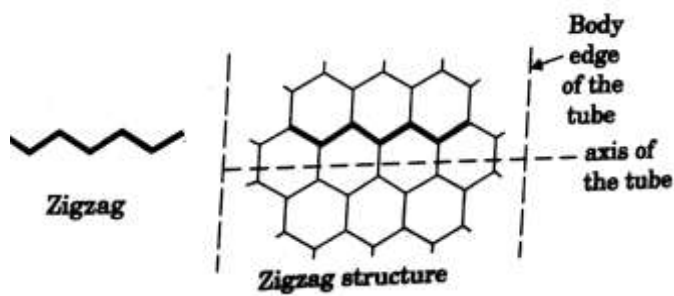
1) ARM CHAIR STRUCTURE:

The axis of the tube is parallel to C-C bonds of carbon hexagons.



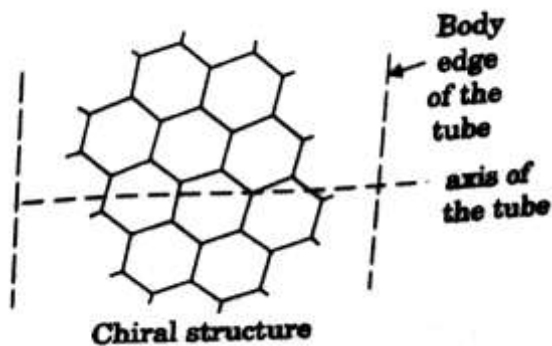
2) ZIG – ZAG STRUCTURE:

The axis of the tube will be perpendicular to C-C bonds.



3) CHIRAL STRUCTURE:

The C-C bond is inclined towards the axis of the tube.



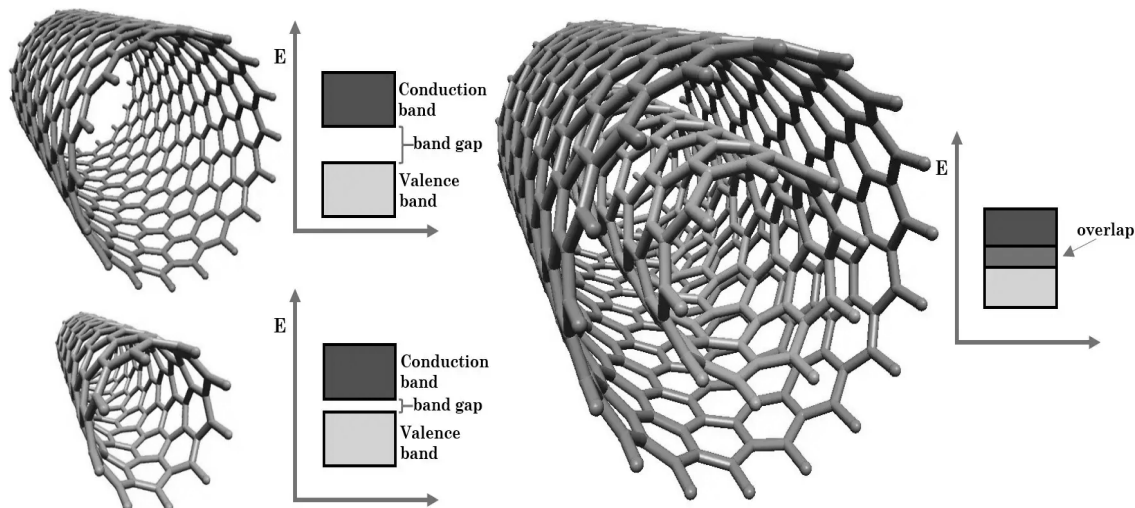
Based on number of layers:

1) SINGLE WALLED NANOTUBES (SWNTs)

SWNT consists of a single graphene cylinder.

2) MULTI WALLED NANOTUBES (MWNTs)

MWNT comprises of several concentric graphene cylinders.



Properties

1. Electrical Properties:

1. Depending on diameter & chirality they behave as metallic or semiconductor.
2. The Energy gap is inversely proportional to diameter of the tube.
3. High electrical conductivity.

2. Mechanical Properties:

1. They have high strength due to C-C bonding.
2. Young's modulus is greater so tensile strength is high compare than steel and Kevlar
3. With stand extreme strain.
4. Highly flexible.

3. Thermal properties

1. Have a low thermal expansion coefficient.
2. It has high thermal conductivity and it increases with decrease in diameter.

4. Chemical properties:

1. Highly resistant to chemical reaction
2. It is difficult to oxidize CNT

5. Physical Properties:

1. They possess high strength to weight ratio.
2. The surface area of nanotube is of the order $10 - 20 \text{ m}^2/\text{g}$ which is higher than graphite.

Applications

- 1) The carbon nanotubes are very light in weight but they are very strong, hence they are used in aerospace.
- 2) They are used in constructing nanoscale electronic devices.
- 3) CNTs are used in battery electrodes, fuel cells, reinforcing fibers etc.
- 4) CNTs are used in the development of flat panel displays for computer monitors and televisions.
- 5) Semiconducting CNTs are used as switching devices.
- 6) Semiconducting CNTs are also used as chemical sensors to detect various gases.

PREPARATION OF CARBON NANO TUBES (CNTs)

Chemical vapor deposition method

Chemical vapor deposition (CVD): CVD is a technique in which the vaporized reactants react chemically and forms a nanomaterial product that is deposited on the substrate.

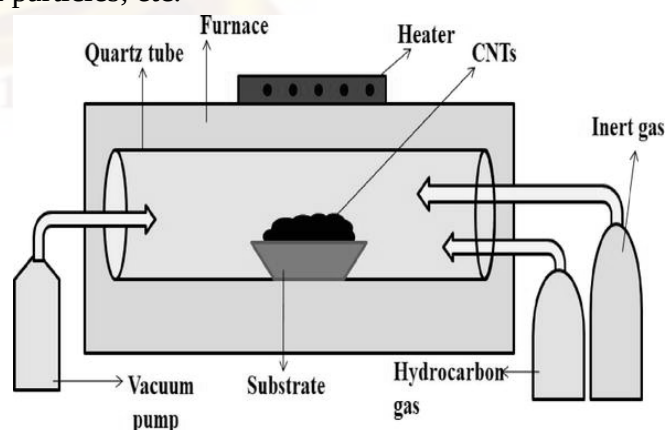
Description and Working

- ❖ A hydrocarbon gas such as acetylene, ethylene, methane, etc can be used as sources for carbon.
- ❖ CNTs are grown on the substrates the commonly used substrates in CVD method are zeolite, silica, silicon plate coated with iron particles, etc.
- ❖ To produce single-walled carbon nanotubes metal catalyst nanoparticles such as iron, cobalt, nickel, molybdenum, iron-molybdenum alloys, etc. are used.
- ❖ Thermal activated CVD which is heated by IR radiation, RF heater, etc.
- ❖ **Conditions maintained:** The following conditions are maintained inside the furnace.

Temperature range: $500-900^\circ\text{C}$.

Inert gas atmosphere: Argon gas.

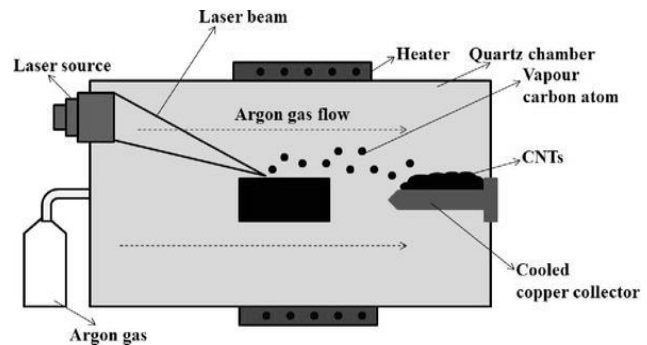
- ❖ CNTs are synthesized by thermal CVD method by using hydrocarbon gas as carbon source. In this method, a quartz tube is placed inside a furnace maintained at high temperature ($500-900^\circ\text{C}$) heated by RF heater.
- ❖ A crucible containing the substrate coated with catalyst nanoparticles is placed inside quartz tube filled with inert gas such as argon gas.
- ❖ The hydrocarbon gas (carbon source) is pumped into the quartz tube which undergoes pyrolysis reaction and forms vapor carbon atoms.



- ❖ These carbon atoms bind to the substrate and join to each other by Vanderwaal force of attraction and grow as multi-walled carbon nanotubes (MWCNTs) on the substrate
- ❖ The obtained CNTs are further purified to get the pure form of CNTs.

Laser Ablation method

- ❖ Laser Ablation method is a Physical Vapor Deposition (PVD) method in which graphite target is vaporized by laser source.
- ❖ In this method the graphite target is placed at the center of quartz chamber filled with argon gas and maintained at 1200°C.
- ❖ The graphite target is vaporized by either continuous laser source or pulsed laser source. The vaporized target atoms (carbon) are swepted toward cooled copper collector by the flow of argon gas.
- ❖ The carbon atoms are deposited and grown as CNTs on cooled copper collector.
- ❖ In case of continuous laser beam, the carbon atoms are continuously vaporized whereas in case of pulsed laser beam the amount of CNTs produced can be monitored as each shot of pulsed laser beam is directly proportional to the amount of carbon atoms vaporized.
- ❖ By this method multi-walled carbon nanotubes are synthesized and to synthesize single-walled carbon nanotubes catalyst nanoparticles of Fe, Co, Ni are used. The obtained CNTs are further purified to get the pure form of CNTs.



5. Quantum Dot Laser

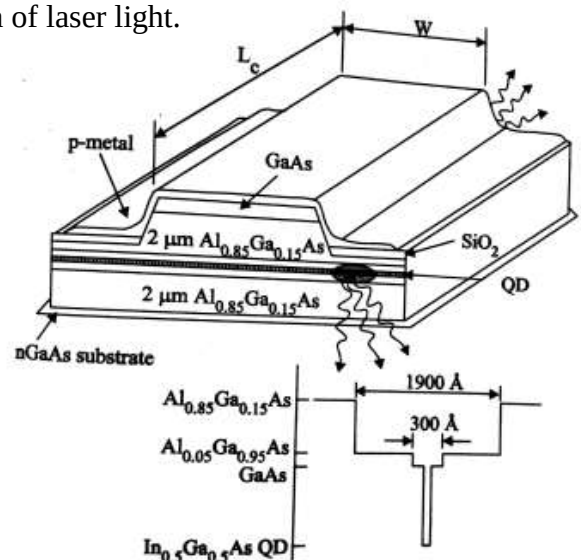
It is a new generation semiconductor laser. It consists of several million nano sized crystals called quantum dots in the active region and they acts as light emitters. i.e., a quantum dot laser is a semiconductor laser that uses quantum dots as the active laser medium in its light emitting region.

Construction

The top p – metal layer has as GsAs contact layer. Immediately below it, there are a pair of two μm thick AlGaAs cladding or bounding layers that surround a 190 nm thick waveguide made up of AlGaAs in between p-metal and n-substrate. Here the waveguide plays a role of conducting the emitted light to the exit ports at the edges of the structure. The waveguide (dark dotted on the figure labeled QD) is a 30 nm thick GaAs region and centered in this region are 12 monolayers of InGaAs quantum dots with a density of $1.5 \times 10^{10} \text{ cm}^{-2}$.

Working

1. The electron – hole recombination causes the emission of laser light.
2. By varying the length and width, the laser light with particular wavelength will be emitted.
3. A particular wavelength of $1.32 \mu\text{m}$ which is in the near infrared region can be produced.
4. The faces of the layer were coated with high reflectively material (ZnSe) where the light is reflected back and forth to increase the stimulated emission and in turn the laser emission is enhanced.



Advantages

1. Broad spectrum with a specific wavelength of light emission can be obtained by changing dot size.
2. It operates even at high frequencies.
3. It is an efficient laser.
4. Gain is 2 to 3 times more than the quantum well.

Disadvantages

1. It is very difficult to form high quality dots (uniform size and higher density)
2. Difficult to manufacture.

Applications

1. They are used in medicine.
2. They are used in display technologies, spectroscopy and telecommunications.
3. They are used in optical transmission system and optical LANs.

